The word salient means projecting. This rotor has a number of projecting poles having their bases riveted to the rotor. It is mainly used in low-speed alternators. The salient 2-pole rotor is shown in the Figure 4.30.

Cylindrical pole rotor

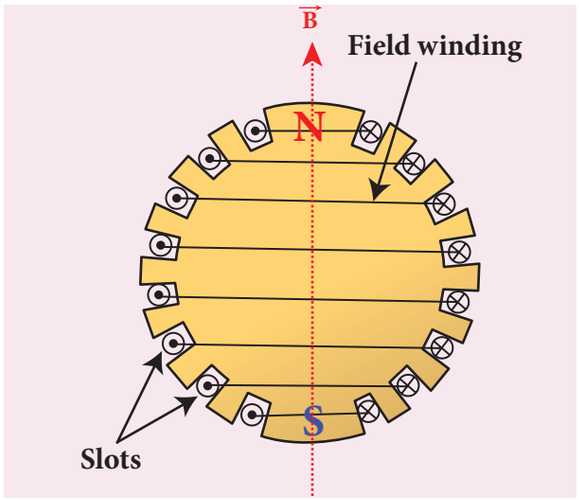


Figure 4.31 Cross-sectional view of cylindrical 2-pole rotor

This rotor consists of a smooth solid cylinder. The slots are cut on the outer surface of the cylinder along its length. It is suitable for very high speed alternators (Figure 4.31).

The frequency of alternating emf induced is directly proportional to the rotor speed. In order to maintain the frequency constant, the rotor must run at a constant speed.

These are standard construction details of alternators. Based on the type of alternator being constructed, the details like number of poles, pole type, number of coils and type of windings would change from one another.

We will discuss the construction and working of two examples, namely single phase and three phase AC generators in the following sections.

4.5.4 Advantages of stationary armature-rotating field alternator

Alternators are generally high current and high voltage machines. The stationary armature-rotating field construction has many advantages. A few of them include:

1) The current is drawn directly from fixed terminals on the stator without the use of brush contacts.
2) The insulation of stationary armature winding is easier.
3) The number of sliding contacts (slip rings) is reduced. Moreover, the sliding contacts are used for low-voltage DC Source.
4) Armature windings can be constructed more rigidly to prevent deformation due to any mechanical stress.

4.5.5 Single phase AC generator

In a single phase AC generator, the armature conductors are connected in series so as to form a single circuit which generates a single-phase alternating emf and hence it is called single-phase alternator.

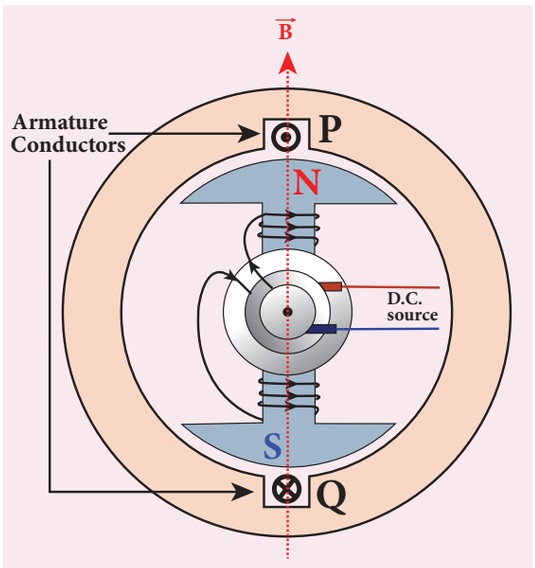


Figure 4.32 Stator core with a rectangular loop and 2-pole rotor

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